

STRUCTURE AND CLASSIFICATION OF NEURONS

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LEARNING OBJECTIVES

- Define nerve fibre.
- Identify parts of a neuron.
- Differentiate b/w unipolar, bipolar & multipolar neuron.
- Differentiate b/w afferent & efferent nerve fibres.
- Describe the structure and function of myelin sheath & nodes of ranvier.
- Differentiate b/w myelinated & non-myelinated nerve fibres.
- Classify nerve fibres on the basis of structure, distribution, origin & type of neurotransmitter.
- Classify nerve fibres on the basis of diameter & conduction velocity (Erlanger-Gasser classification)

INTRODUCTION

The nervous system is the chief controlling and co-ordinating system of the body. It regulates and integrates all body activities by transmitting electrical and chemical signals.

Nervous system receives information(stimuli) from inside and outside the body , process information and controls body actions accordingly. Information travels in the form of impulse.

The nervous system consists of:

- 1) Brain
- 2) Spinal cord
- 3) Nerves.

The nervous system is a complex network of specialized cells that co-ordinates body functions by transmitting impulses.

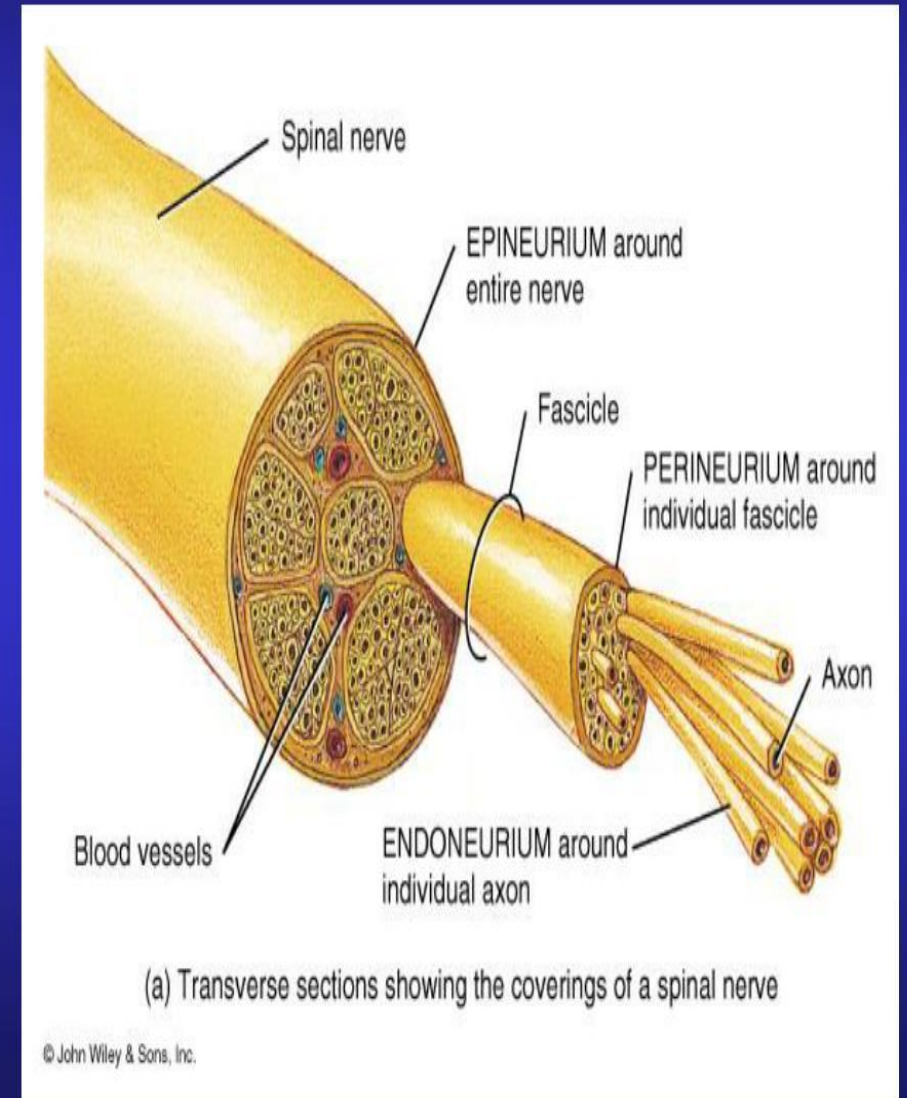
ORGANIZATION OF NERVE

Each nerve is formed by many bundles or groups of nerve fibres. Each bundle or group of nerve fibres is called a fasciculus. Each nerve fibre is actually the axon part of neuron.

COVERING OF NERVE

The whole nerve is covered by tubular sheath. This sheath is called *epineurium*. Each fasciculus is covered with *perineurium* and each nerve fibre is covered by *endoneurium*.

Organization of a Nerve of the PNS



NEURON

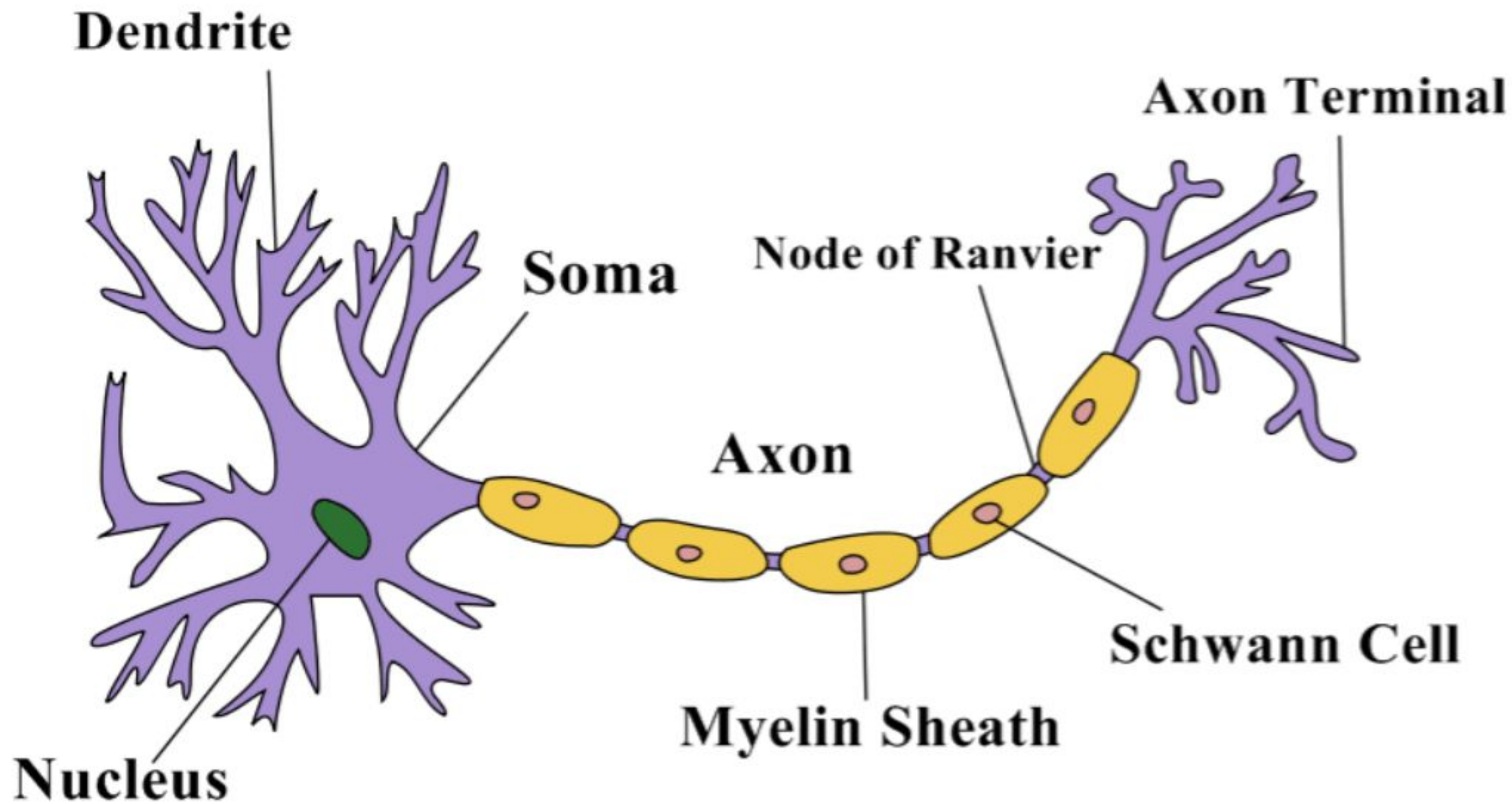
A neuron is a structural and functional unit of nervous system.

- It is a specialized cell designed to receive, process and transmit electrical and chemical signals in the body.

Structure of a neuron

A neuron consists of:

- 1) Dendrites
- 2) Cell body(soma)
- 3) Axon
- 4) Axon terminal



1. CELL BODY(soma)

Nerve cell body also known as *soma* or *perikaryon* is irregular in shape. Like any other cell it is constituted by a mass of cytoplasm called neuroplasm. The neuroplasm contains a large nucleus, nissl bodies, neurofibrils, mitochondria & golgi apparatus.

NUCLEUS

Each neuron has a single nucleus centrally placed in the nerve cell body. Nucleus has one or two prominent nucleoli. Nucleus of a neuron does not contain centrosome so the nerve cell cannot multiply like other cells.

NISSL BODIES

Nissl bodies are small basophilic granules found in the cytoplasm of neurons. Nissl bodies are membranous organelles having ribosomes so these bodies are concerned with the synthesis of proteins in the neurons. These bodies are present in the soma and dendrite but not in axon and axon hillock. Proteins formed in the soma are then transported to the axon by axonal flow. Nissl bodies are called *tigroid substances* because these bodies are responsible for tigroid or spotted appearance of soma after staining.

NEUROFIBRILS

Neurofibrils are thread-like structures present in the form of network in the soma and the nerve processes. Presence of neurofibrils is another characteristic feature of neurons. The neurofibrils consists of microfilaments and microtubules.

MITOCHONDRIA

Mitochondria are present in soma & in axon. Like other cells, these are the *power house* of nerve cell responsible for the production of energy(ATP).

GOLGI APPARATUS

Like other cells, golgi apparatus of nerve cells are concerned with the processing and packaging of proteins into granules.

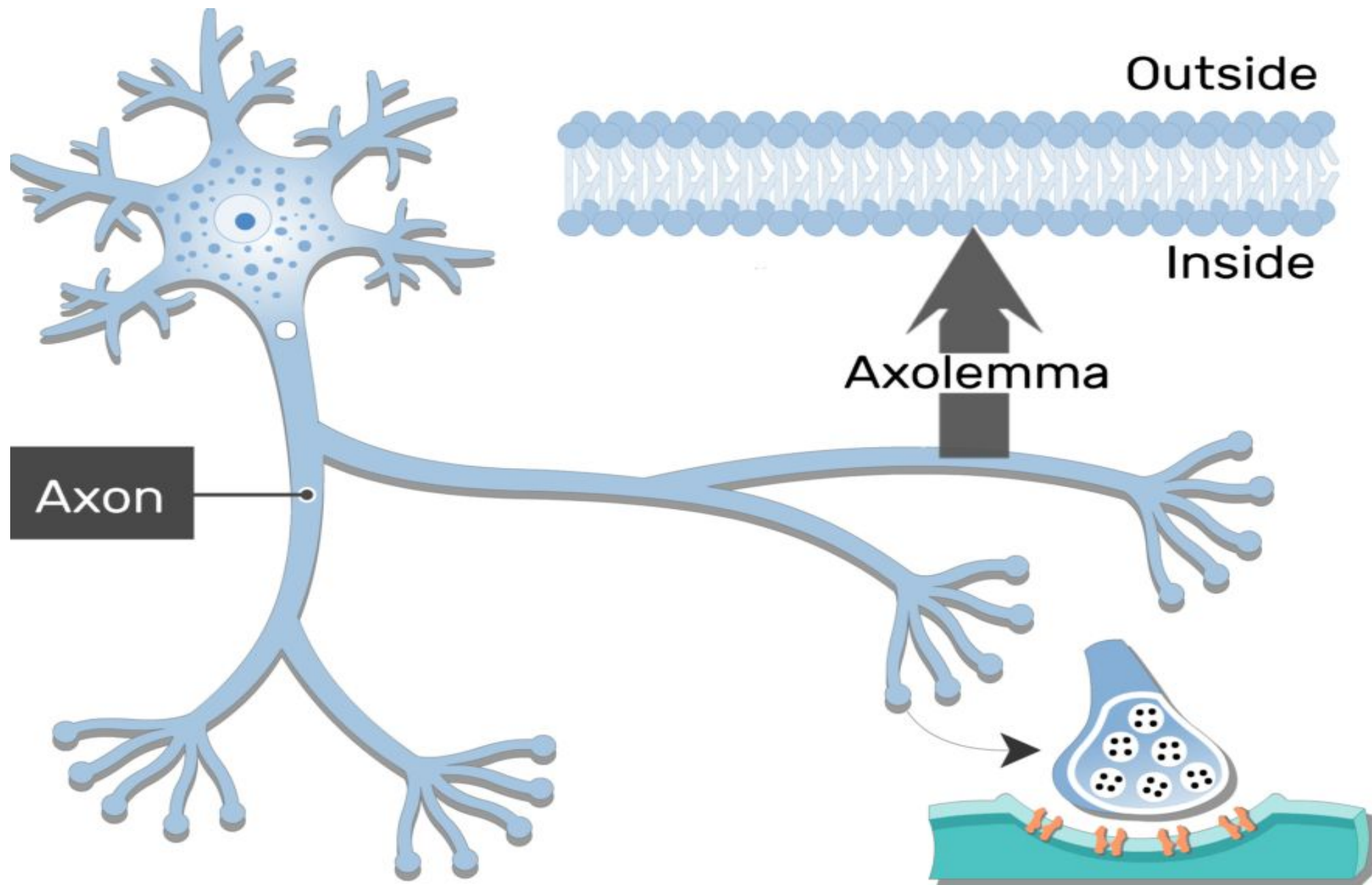
2. DENDRITE

Dendrites are short, branched extensions that arise from the cell body of neurons. Dendrites contain receptor proteins on their surface. Small protrusions called *dendritic spines* are present on the surface of dendrites to increase the surface area. Dendrites receive and transmit impulses towards the cell body. Usually the dendrites are shorter than axon.

3. AXON

Axon is a long single projection that carries impulses away from the cell body. Axon consists of:

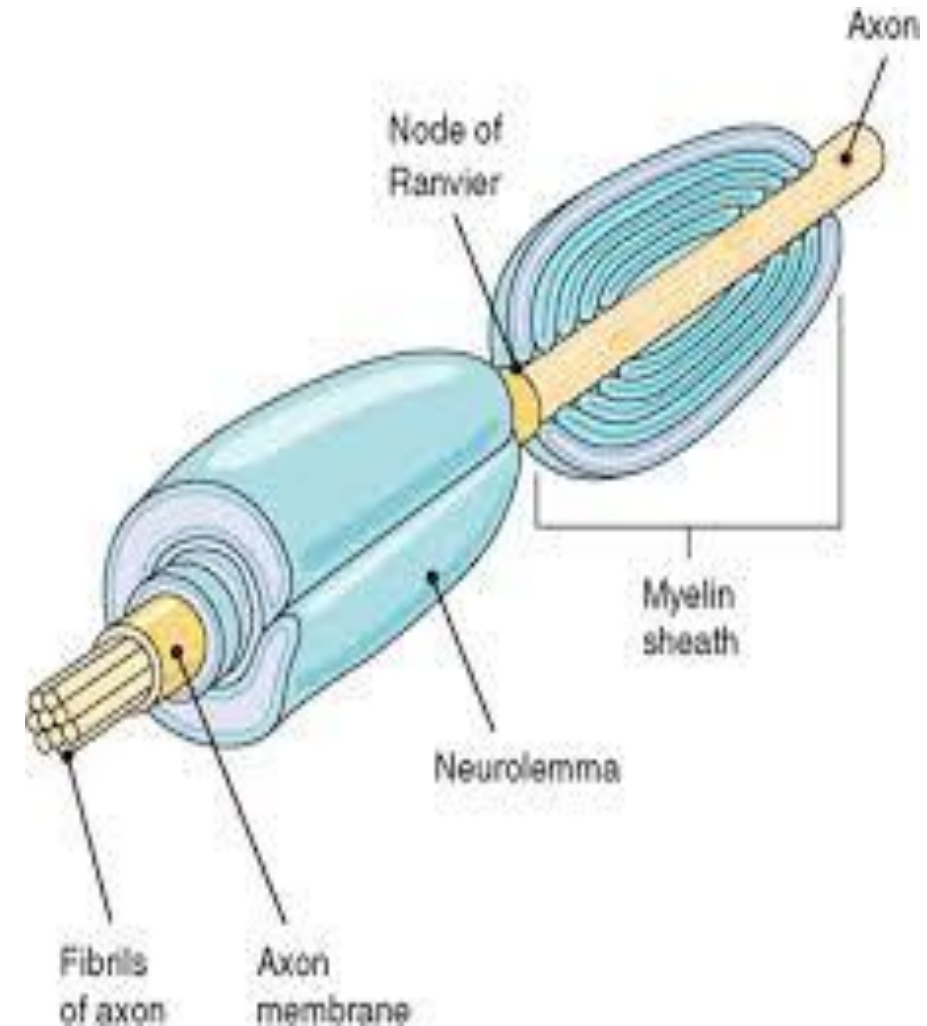
- **AXON HILLOCK:** Cone shaped region where the axon begins. Axon hillock is the site for the initiation of action potential.
- **AXON PROPER:** Axon fibre is a long cylindrical fibre of a neuron. It may vary in length from a few millimeters (like in sensory neuron) to over a meter (like in motor neurons).



INTERNAL STRUCTURE OF AXON

Axon has a long central core of cytoplasm called **axoplasm**. Axoplasm is covered by a tubular sheath like membrane called **axolemma**. Axolemma is the continuation of cell membrane of nerve cell body. Axoplasm along with axolemma is called the **axis cylinder** of the nerve fibre. Axoplasm contains mitochondria, neurofibrils and axoplasmic vesicles. Because of the absence of nissl-bodies in the axon, proteins necessary for the nerve fibres are synthesized in the soma and transported from soma to axon by **axonal flow**. Some neurotransmitter substances are also transported by axonal flow from soma to axon.

Axis cylinder of the nerve fibre is covered by a membrane called **neurolemma**.



MYELIN SHEATH

Myelin sheath is a thick lipoprotein sheath that insulates the myelinated nerve fibres. Myelin sheath is not a continuous sheath, it is absent at regular intervals. The area where myelin sheath is absent is called **nodes of renvier**. Segment of nerve fibre b/w two nodes is called **internode**. Myelin sheath is formed by **schwann cells** in peripheral nervous system and by **oligodendrocytes** in central nervous system. Formation of myelin sheath around the axon of myelinated nerve fibres is called **myelinogenesis**. Myelin sheath is responsible for white color of nerve fibres. On the basis of presence or absence of myelin sheath nerve fibres are named as

- **Myelinated nerve fibres:** nerve fibres that are insulated by myelin sheath
- **Non-myelinated nerve fibres:** nerve fibres that are not covered by myelin sheath.

FUNCTIONS OF MYELIN SHEATH

Myelin sheath is responsible for faster conduction of impulse through the nerve fibres. In myelinated nerve fibres the impulses jump from one node to another node. This type of impulse transmission is called *saltatory conduction*.

Myelin sheath has an insulating capacity so the myelin sheath restricts the impulse within single nerve fibre and prevents the stimulation of neighbouring nerve fibre.

AXON TERMINALS(synaptic knobs)

These are the branched endings of axon. It contains synaptic vesicles filled with neurotransmitters. Axon terminals have mitochondria for energy supply.

Axon terminals release neurotransmitter into the synaptic cleft. They also communicate with another neuron, muscle fibre or gland.

SYNAPSE

Synapse is the junction b/w two neurons. It consists of *presynaptic terminal*(axon terminal), *synaptic cleft*(small gap b/w two neurons) and *post synaptic terminal*(usually dendrite of next neuron).

CLASSIFICATION OF NEURONS

Neurons are classified by three different methods:

- 1) Depending upon the number of poles
- 2) Depending upon the function
- 3) Depending upon the length of axon
- 4) Depending upon the structure
- 5) Depending upon the distribution of nerve fibres
- 6) Depending upon the origin of nerve fibres
- 7) Depending upon the secretion of neurotransmitter
- 8) Depending upon the diameter and conduction velocity of impulse

9) *DEPENDING UPON NO. OF POLES*

Based on the number of poles from which the nerve fibre arise, neurons are classified into three types.

- Unipolar neurons
- Bipolar neurons
- Multipolar neurons

- **UNIPOLAR NEURONS**

Unipolar neurons are the neurons that have only one pole. From a single pole, both axon and dendrites arise.

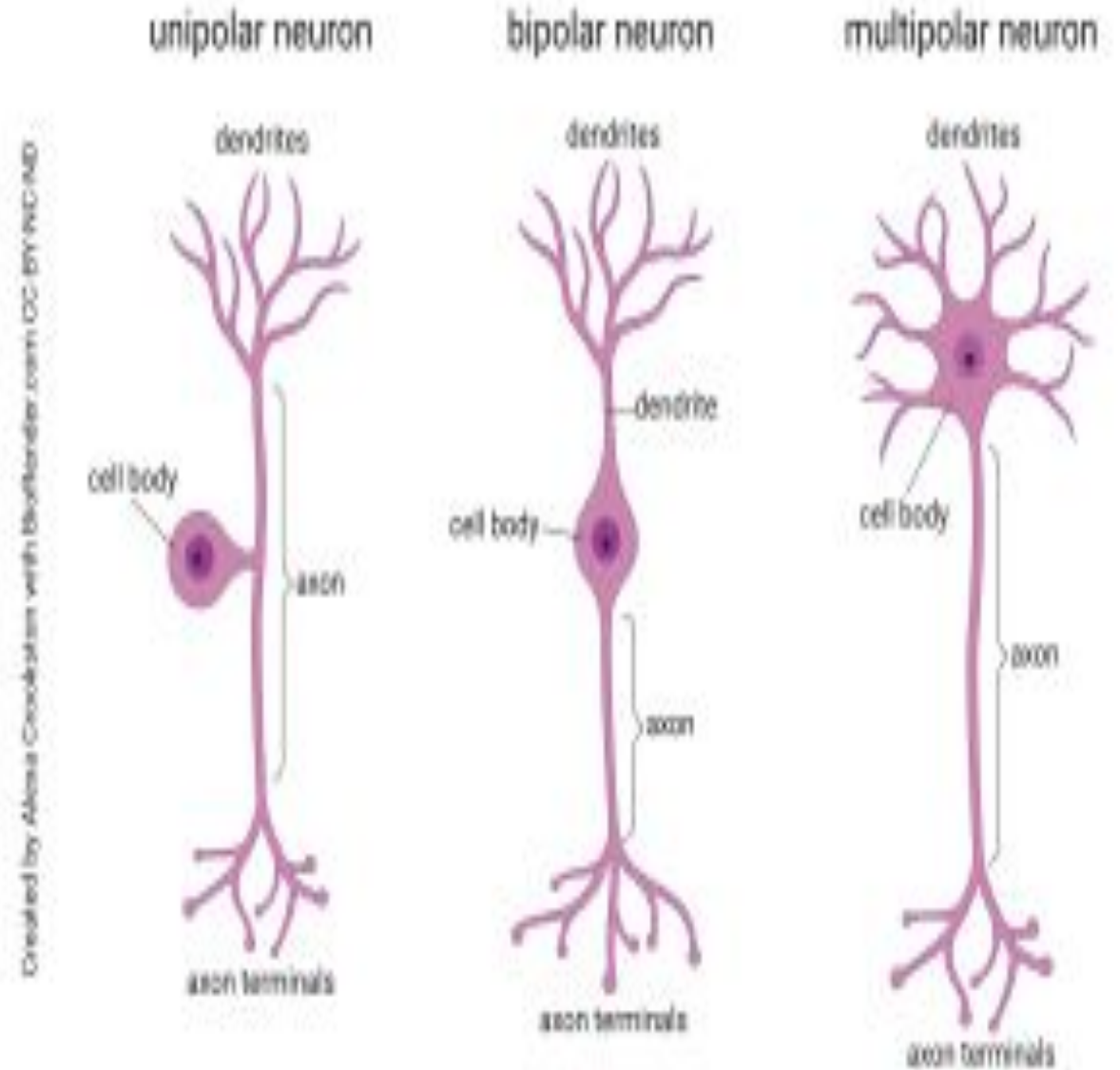
This type of nerve cells present only in embryonic stage in human beings.

- **BIPOLAR NEURONS**

Neurons with two poles are called bipolar neurons. Axon arise from one pole and dendrites arise from the other pole.

- **MULTIPOLAR NEURONS**

Neurons with many poles are called multipolar neurons. One pole gives rise to axon and all other poles gives rise to dendrites.



2) DEPENDING UPON THE FUNCTION

On the basis of function, nerve cells are classified into two types:

- Motor or efferent neurons
- Sensory or afferent neurons
- *MOTOR OR EFFERENT NEURONS*

Neurons which carry the motor impulses from central nervous system to peripheral effector organs like muscles, glands, blood vessels etc. Generally motor neuron has a long axon and short dendrites.

- *SENSORY OR AFFERENT NEURONS*

Sensory or afferent neurons carry the sensory impulses from periphery to the central nervous system. Generally sensory neurons have a short axon and long dendrites.

3) DEPENDING UPON THE LENGTH OF AXON

Depending upon the length of axon, neurons are divided into two types:

- Golgi type I neurons
- Golgi type II neurons
- *GOLGI TYPE I NEURONS*

Golgi type I neurons have long axons. Cell body of these neurons is in different part of central nervous system and their axons reach the remote peripheral organs.

- *GOLGI TYPE II NEURONS*

These neurons have short axons. These neurons are present in central cortex and spinal cord.

4) DEPENDING UPON THE STRUCTURE

Based upon structure, nerve fibres are classified into two types:

- Myelinated nerve fibres
- Non-myelinated nerve fibres
- *MYELINATED NERVE FIBRES*

Myelinated nerve fibres are covered with myelin sheath.

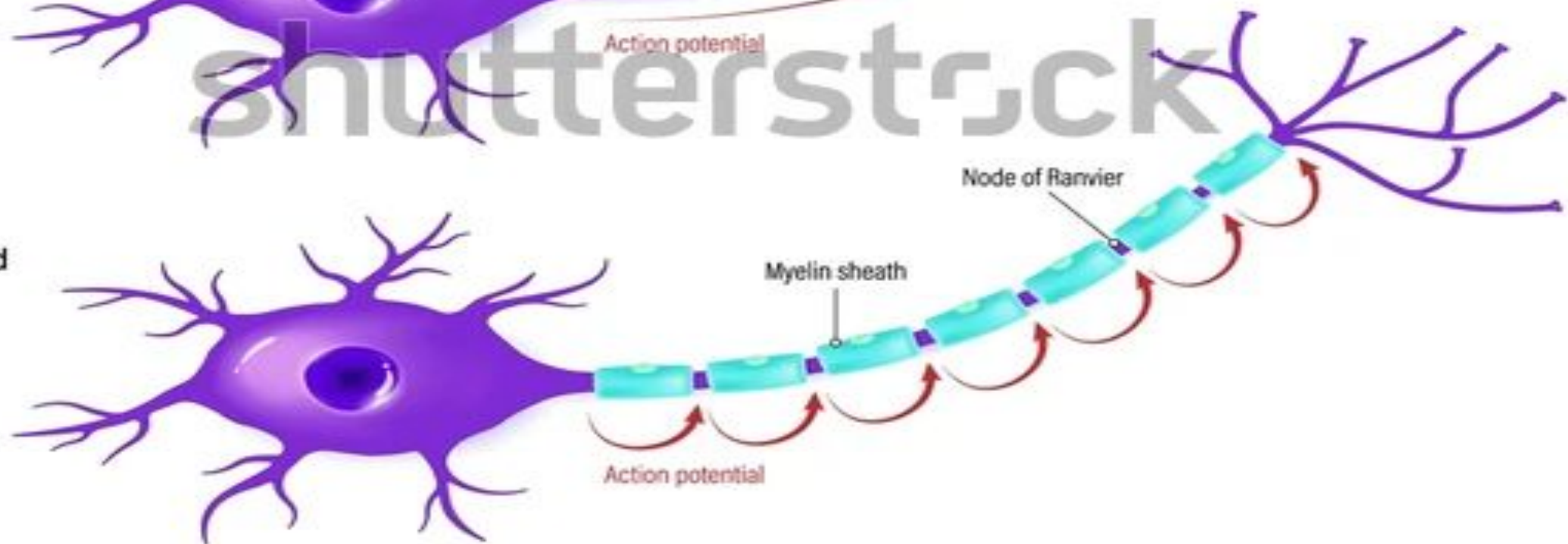
- *NON-MYELINATED NERVE FIBRES*

Non-myelinated nerve fibres are the nerve fibres that are not covered by myelin sheath.

Unmyelinated
Axon



Myelinated
Axon



5) DEPENDING UPON THE DISTRIBUTION

Nerve fibres are classified into two types on the basis of distribution:

- Somatic nerve fibres
- Visceral or autonomic nerve fibres
- *SOMATIC NERVE FIBRES*

Somatic nerve fibres supply the skeletal muscles of the body.

- *AUTONOMIC OR VISCERAL NERVE FIBRES*

Autonomic nerve fibres supply the various internal organs of the body.

6) DEPENDING UPON THE ORIGIN

On the basis of origin, nerve fibres are divided into two types:

- Cranial nerve fibres
- Spinal nerve fibres
- *CRANIAL NERVE FIBRES*

Nerve fibres that arise from the brain are called cranial nerve fibres.

- *SPINAL NERVE FIBRES*

Nerve fibres that arise from spinal cord are called spinal nerve fibres.

7) DEPENDING UPON THE SECRETION OF NEUROTRANSMITTER

Depending upon the neurotransmitter substance secreted, nerve fibres are divided into two types:

- Adrenergic nerve fibres
- Cholinergic nerve fibres
- *ADRENERGIC NERVE FIBRES*

These nerve fibres secrete noradrenaline.

- *CHOLINERGIC NERVE FIBRES*

Cholinergic nerve fibres secrete acetylcholine.

8) DEPENDING UPON DIAMETER & VELOCITY OF IMPULSE CONDUCTION

Erlanger and Gasser classified the nerve fibres into three major types:

- Type A nerve-fibres
- Type B nerve-fibres
- Type C nerve-fibres

Among these nerve fibres, type A nerve fibres are the thickest fibres and type C fibres are the thinnest nerve fibres. Type C fibres are also known as type IV fibres.

Except type C fibres, all the nerve fibres are myelinated.

SUBDIVISIONS OF TYPE A NERVE FIBRES

Type A nerve fibres are divided into four types:

- 1) Type A alpha or type I nerve fibres
- 2) Type A beta or type II nerve fibres
- 3) Type A gamma nerve fibres
- 4) Type A delta or type III nerve fibres.

VELOCITY OF IMPULSE CONDUCTION

Velocity of impulse through nerve fibre is directly proportional to the thickness of nerve fibre.

Erlanger gasser classification

Fiber type	Functions	Diameter-mm	Conduction Velocity/M/se
A- α	Proprioception somatic motor	12 - 20	70 - 120
A- β	Touch, Pressure	5 - 12	30-70
A- γ	Motor to spindle	3 - 6	15 - 30
A- δ	Pain,tem,touch	2 - 5	12 - 30
B	Preganglionic autonomic	3	3 - 15
C	Dorsal root pain-reflex	0.4-1.2	0.5-2
	Sympathetic post gang	0.3-1.3	0.7-2.3

POST ASSESSMENT

Q1. Insulating layer around the axon is called:

- a) Axoplasm
- b) Nissl body
- c) Myelin sheath
- d) Neurolemma

Q2. Myelin sheath in PNS is produced by:

- a) Oligodendrocytes
- b) Astrocytes
- c) Schwann cells
- d) Microglia

Cont..

Q3. Gaps b/w myelin segments are called:

- a) Dendrites
- b) Axon hillock
- c) Synapse
- d) Nodes of ranvier

Q4. Dendrites carry impulses away from the cell body? (T/F)

Q5. Axon hillock is the site of initiation of action potential? (T/F)

The background of the image is a dense, repeating pattern of dark blue, glossy leaves. The leaves are elongated and pointed, with visible veins. They are arranged in a way that creates a textured, almost three-dimensional effect. The lighting is soft, highlighting the edges and surfaces of the leaves.

Thank you